



Faculty of Mathematics and Computer Science

Knowledge Discovery (KD)

Steel Industry Knowledge Discovery on Continuous Casting Machine Dataset

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Abstract

This research addresses **predictive maintenance and process optimization in steel manufacturing**, focusing on the continuous casting stage. A **Continuous Casting Machine (CCM)** transforms molten steel into semi finished billets. A key component is the water cooled copper mold **sleeve**, which initiates solidification. Due to high thermal stress and friction, the sleeve degrades over time, making its monitoring important for reducing downtime and maintenance costs.

The main objective is to analyse the **Remaining Useful Life (RUL)** of mold sleeves using industrial data. A preprocessing pipeline was applied to correct this issue by unifying time and date into a single timeline and ordering records by sleeve identifier. A new **RUL** definition was introduced based on total processed steel, ensuring a monotonic and physically consistent measure.

For analysis, numerical variables were discretized into categorical levels (*Low, Medium, High*). The study combines machine learning and knowledge discovery methods. Supervised models, including **Random Forest Regressor (RFR)** and **Random Forest Classifier (RFC)**, were used for baseline prediction and feature importance. Unsupervised methods, including **DBSCAN**, **PCA**, and **Isolation Forest (IF)**, were applied for structure discovery and anomaly detection.

The main knowledge discovery method is **Formal Concept Analysis (FCA)**, which organizes data into concept lattices representing relationships between process variables and equipment states. Association rule mining (Apriori algorithm) was also applied to extract logical dependencies. The analysis was supported by scaling methods implemented using the **ToscanaJ Suite**.

Advanced extensions include **Triadic Formal Concept Analysis (3FCA)**, which introduces operating conditions as a third dimension, enabling analysis of context-dependent relationships. **Temporal Formal Concept Analysis (TCA)** incorporates time to model state transitions and reconstruct lifecycle trajectories.

Overall, combining machine learning with formal concept analysis enables structured interpretation of industrial data and improves understanding of mold sleeve degradation.

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Keywords: (KD) Knowledge Discovery; (KDD) Knowledge Discovery in Databases; (AI) Artificial Intelligence; (ML) Machine Learning; (SL) Supervised Learning; (UL) Unsupervised Learning; (DA) Data Analysis; (CCM) Continuous Casting Machine; (SI) Steel Industry; (SE) Software Engineer; (DM) Data Mining; (FCA) Formal Concept Analysis; (TCA) Temporal Concept Analysis; (RUL) Remaining Useful Life; (RFR) Random Forest Regressor; (RFC) Random Forest Classifier; (DBSCAN) Density-Based Spatial Clustering of Applications with Noise; (PCA) Principal Component Analysis; (IF) Isolation Forest; (AE) Attribute Exploration; (3FCA) Triadic Formal Concept Analysis; (CTS) Conceptual Time System

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1. Introduction

This research paper is driven by my interest in the **Steel Industry (SI)**, more specifically in the **continuous casting process**, which represents one of the most critical stages in modern steel production.

My motivation for selecting this topic is strongly connected to my **professional career**, as I currently work as a **Software Engineer (SE)** in the **industrial automation sector**, specializing in **Industrial Furnaces**. These furnaces operate in a downstream stage of the steel production chain, processing billets and semi-finished products that originate from the **Continuous Casting Machine (CCM)**.

Although my daily work focuses on thermal treatment, reheating and process automation within industrial furnace systems, **understanding the complete lifecycle of steel production is essential for developing a broader and more integrated perspective of the manufacturing process**. The quality of the material entering industrial furnaces is directly influenced by the parameters and operating conditions of the continuous casting process. Defects, inaccuracies, thermal stresses and material inconsistencies produced during casting can propagate through subsequent production stages, ultimately affecting furnace operation and overall plant efficiency.

Studying the **CCM** and the factors that influence its performance is particularly relevant to **my professional development**. From an academic perspective, the selected dataset provides an excellent opportunity to apply techniques from **Artificial Intelligence (AI)/Machine Learning (ML)**, **Knowledge Discovery in Databases (KDD)**, **Data Mining (DM)**, **Formal Concept Analysis (FCA)** and **Temporal Concept Analysis (TCA)**. The dataset contains real industrial measurements related to steel chemistry, casting, cooling and equipment wear, making it suitable for extracting hidden patterns and identifying relationships between variables that can support decision making in industrial environments.

2. Literature Review

The analysis performed in this research combines several disciplines previously mentioned from the fields of **artificial intelligence** and **knowledge discovery**. *Artificial Intelligence (AI)* [16] provides the theoretical framework for developing systems capable of performing tasks that typically require human intelligence, while *Machine Learning (ML)* [3] focuses on constructing models that learn patterns from historical data and make predictions on unseen observations.

Knowledge Discovery in Databases (KDD) [7] represents the overall process of extracting useful, valid, and understandable knowledge from large datasets. *Data Mining (DM)* [10] is considered a core step within the **KDD** process and involves the application of statistical and machine learning techniques to identify patterns, correlations and predictive relationships in data.

To obtain interpretable knowledge from industrial process data, this work also employs *Formal Concept Analysis (FCA)* [8], a mathematical framework for discovering conceptual structures and relationships between objects and attributes. **FCA** enables the construction of **concept lattices**.

Furthermore, *Temporal Concept Analysis (TCA)* [20] extends **FCA** by **incorporating temporal information and state transitions**. This approach allows the investigation of system evolution over time.

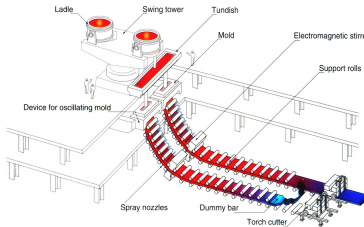


Fig. 1: Image from "Continuous casting practices for steel: Past, present and future" [9]

3. Dataset

3.1. Description and Selection

The dataset used throughout this research was obtained from the **Kaggle** repository *RUL Dataset from Continuous Casting Machine*, published by Yuriy Katser [13].

According to the authors, the dataset was collected from a **real industrial steel production facility** and was specifically prepared for the study of **Remaining Useful Life (RUL)** prediction in continuous casting machine mould sleeves. Before publication, the original production data were processed, cleaned, aggregated and structured to support **ML** and predictive maintenance applications.

Continuous Casting Machine (**CCM**) is an **industrial installation that transforms molten steel into solid billets of a predefined cross-section**, which are subsequently processed through rolling mills to produce finished steel products. One of the most critical components of the **CCM** is the **mould sleeve**, also referred to as the crystallizer sleeve, a water-cooled copper tube responsible for initiating the solidification process. As molten steel comes into contact with the sleeve walls, a primary solid shell is formed, allowing the billet to maintain its geometry while moving through the casting process.

The operational degradation of mould sleeves represents a significant industrial challenge. Defects can propagate through downstream manufacturing stages and negatively affect rolling quality, production efficiency and operational costs. Mould sleeves service life is approximately 17,000 tons for billets of section (180 * 180) mm and 13,000 tons for billets of section (150 * 150) mm. Premature sleeve replacement increases maintenance costs, equipment downtime and operational risks.

A particularly important attribute of the dataset is the **RUL** variable, which value is derived from the *resistance, tonn* attribute by calculating, for each unique combination of sleeve, crystallizer and casting stream, the difference between the maximum recorded resistance (corresponding to sleeve failure) and the current resistance value. Consequently, the **RUL** attribute represents the remaining production capacity of the mould sleeve before replacement becomes necessary.

The wide range of the available attributes makes the dataset particularly appropriate for *Formal Concept Analysis (FCA)*, *Temporal Concept Analysis (TCA)* and the construction of conceptual scales within **ToscanaJ**, for both predictive and explainable knowledge extraction.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
date	workpiece	steel_type	doc	requi	cast_in_ro	workpiece	alloy_type	steel_weight	slag_weight	metal_resl	steel_v	grab1_nur	steel_temj	grab2_nur	resistance	swing	freq	crystallize	alloy_spee	water_con	water_tem	water_con	water_con	water_g
1/5/2020	144.9	Arm240	DOC 3402		4	150x150	open	145.3	1.8	0.4	145	4		5	3316	180	9	3	1950	9	360	400	1	
1/5/2020	165.9	St3sp	Contract		10	150x150	open	166.3	1.8	0.4	166	7	1563	5	4955	180	10	3	1950	9	360	420	1	
1/5/2020	168	Arm240	DOC 3402		5	150x150	open	168.4	1.8	0.4	168	7	1579	5	3345	180	10	3	1950	9	340	400	1	
1/5/2020	170.1	St3sp	Contract		7	150x150	open	170.5	1.8	0.4	170	12	1563	5	3400	180	10	3	1950	9	360	420	1	
1/5/2020	163.8	St3sp	Contract		12	150x150	open	164.2	1.8	0.4	164	12	1565	5	3536	180	10	3	1950	9	360	420	1	
1/5/2020	163.8	St3sp	Contract		14	150x150	open	164.2	1.8	0.4	164	4	1568	5	3591	180	10	3	1950	9	360	420	1	
1/5/2020	163.8	Arm240	DOC 3402		5	150x150	open	164.2	1.8	0.4	164	7	1573	3	3123	165	10	3	1950	9	320	400	1	
1/5/2020	107.428	Arm240	DOC 3402		2	150x150	close	135.828	1.8	0.4	135		1579	4	3209	180	7	2	1950	9	200	250	1	
1/5/2020	161.7	St3sp	Contract		6	150x150	open	162.1	1.8	0.4	162	3	1568	5	3373	180	10	3	1950	9	360	410	1	
1/5/2020	165.9	St3sp	Contract		13	150x150	open	166.3	1.8	0.4	166	5	1566	5	3564	180	10	3	1950	9	360	420	1	
1/5/2020	174.3	St3sp	Contract		15	150x150	open	174.7	1.8	0.4	174	7	1566	5	3620	180	10	3	1950	9	360	420	1	
1/5/2020	165.9	St3sp	Contract		8	150x150	open	166.3	1.8	0.4	166	5	1564	5	3427	180	10	3	1950	9	360	420	1	
1/5/2020	165.9	St3sp	Contract		9	150x150	open	166.3	1.8	0.4	166	5	1565	5	3454	180	10	3	1950	9	360	420	1	
1/5/2020	165.9	St3sp	Contract		10	150x150	open	166.3	1.8	0.4	166	7	1563	5	3482	180	10	3	1950	9	360	420	1	
1/5/2020	168	St3sp	Contract		11	150x150	open	168.4	1.8	0.4	168	3	1564	5	3509	180	10	3	1950	9	360	420	1	
1/5/2020	157.449	Arm240	DOC 3402		1	150x150	close	158.449	1.8	0.4	158	12	1584	4	3184	165	10	3	1950	8	210	250	1	
1/5/2020	159.6	Arm240	DOC 3402		1	150x150	open	160.6	1.8	0.4	160	3	1583	5	3236	180	9	2	1950	9	260	320	1	
1/5/2020	170.1	Arm240	DOC 3402		2	150x150	open	170.5	1.8	0.4	170	12	1582	5	3265	180	9	2	1950	9	320	350	1	
1/5/2020	163.8	Arm240	DOC 3402		3	150x150	open	164.2	1.8	0.4	164	5	1572	5	3293	180	9	3	1950	9	350	400	1	
1/5/2020	168	St3sp	Contract		16	150x150	open	168.4	1.8	0.4	168	3	1565	5	3646	180	10	3	1950	9	360	420	1	
1/5/2020	170.1	St3sp	Contract		17	150x150	open	170.5	1.8	0.4	170	12	1566	5	3673	180	10	3	1950	9	360	420	1	
1/5/2020	163.8	St3sp	Contract		18	150x150	open	164.2	1.8	0.4	164	5	1565	5	3700	180	10	3	1950	9	360	420	1	
1/5/2020	157.449	Arm240	DOC 3402		1	150x150	close	158.449	1.8	0.4	158	12	1584	4		165	10	3	1950	8	210	250	1	
1/5/2020	165.9	St3sp	Contract		9	150x150	open	166.3	1.8	0.4	166	5	1565	5	4928	180	10	3	1950	9	360	420	1	

Fig. 2: Dataset Excel Visualization

3.2. Dataset Attributes

Table 1 summarizes the main categories of attributes contained in the dataset.

Table 1: Main dataset attributes and their descriptions

Attribute(s)	Description
date	Date of the casting operation
time_temperature_measurement1, time_temperature_measurement2, sample_time_continuous_caster	Time-related process measurements
workpiece_weight, steel_weight_theoretical, steel_weight	Actual and theoretical steel mass processed during a cast
steel_type	Steel grade produced during the casting process
doc_requirement	Production specification or quality requirement associated with the cast
cast_in_row	Sequential number of the cast within a production campaign
workpiece_slice_geometry	Cross-sectional geometry of the produced billet (e.g., 150×150 mm)
alloy_type	Alloy production mode or casting configuration
slag_weight_close_grab1, metal_residue_grab1, residuals_grab2, technical_trim	Process by-products and material losses generated during casting
grab1_num, grab2_num	Sampling or process operation identifiers
steel_temperature_grab1, temperature_measurement1, temperature_measurement2	Steel temperature measurements recorded at different process stages
resistance	Accumulated tonnage processed by the mould sleeve, representing equipment wear
swing_frequency	Oscillation frequency of the crystallizer during casting
crystallizer_movement	Oscillation amplitude or movement of the crystallizer
alloy_speed	Casting speed of the molten steel
water_consumption, water_temperature_delta	Primary cooling system operating parameters
water_consumption_secondary cooling_zone_1–3	Cooling water flow rates in the secondary cooling zones
kind, quantity	Production batch characteristics and associated quantities
Ce, C, Si, Mn, S, P, Cr, Ni, Cu, As, Mo, Nb, Sn, Ti, V, Al, Ca, N, Pb, Mg, Zn	Chemical composition of the steel expressed as elemental percentages
sleeve	Unique identifier of the mould sleeve
num_crystallizer	Identifier of the crystallizer used during casting
num_stream	Casting strand identifier
RUL	RUL of the mould sleeve, expressed in tons of steel production remaining before replacement. This attribute represents the target variable used throughout the analysis

3.3. Dataset Analysis

The original **RUL** attribute provided in the dataset **was found to be difficult to interpret** as a true remaining useful life measure. During the exploratory analysis, several sleeves exhibited multiple **RUL** countdown sequences, where the **RUL** value reached zero and later increased again. This behaviour is inconsistent with the expected lifecycle of a physical component, since a sleeve cannot regain useful life after it has been consumed. According to the dataset documentation, the original **RUL** was derived from resistance measurements rather than from the actual cumulative usage of a sleeve. Therefore, a new **RUL** metric was **calculated based on the total amount of steel processed by each sleeve over time**. This produced a monotonically decreasing value that better represents the progressive wear of the sleeve throughout its operational lifecycle.

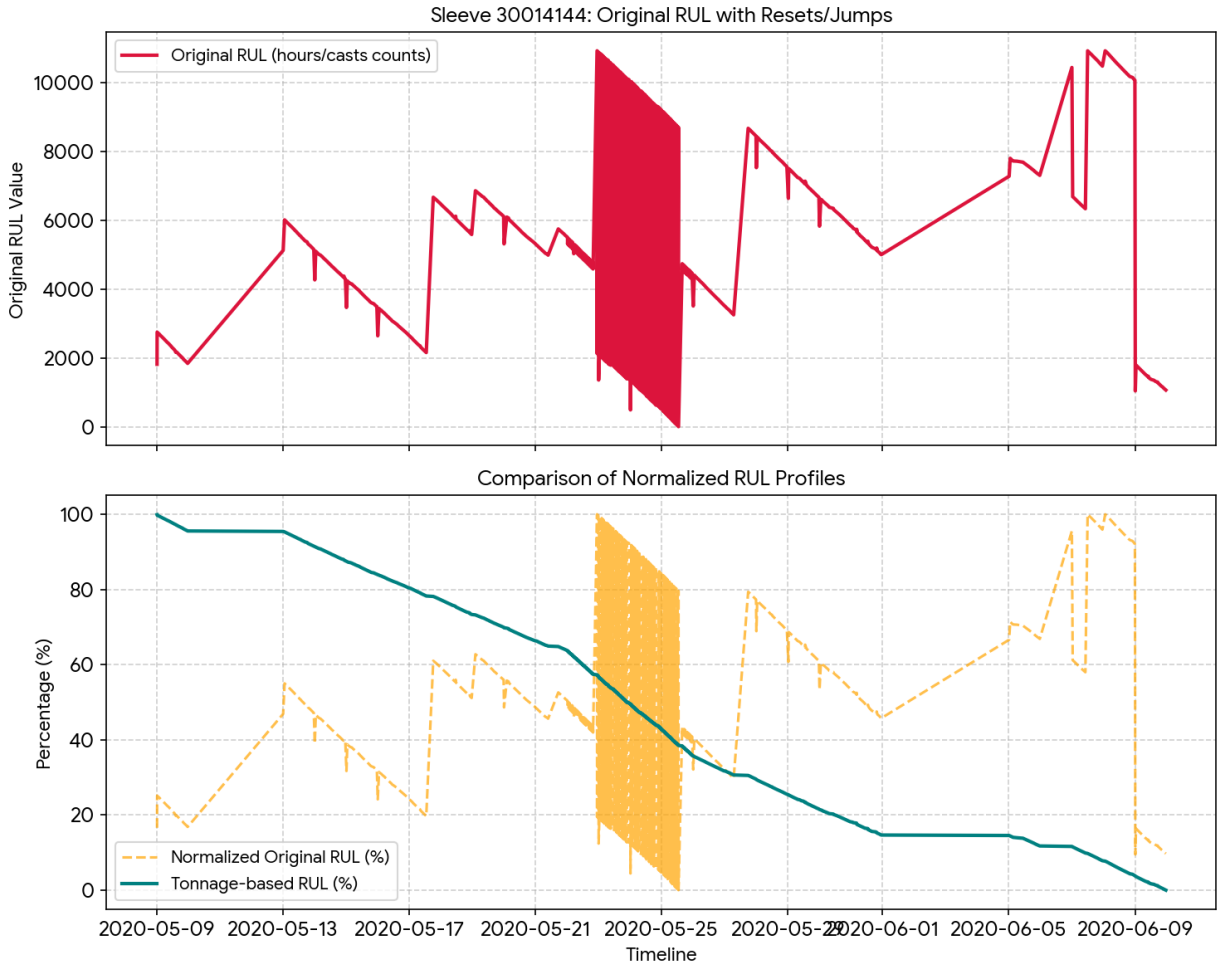


Fig. 3: Original **RUL** unclarity over a normal **RUL**

3.4. Dataset Preprocessing

The preprocessing stage was designed to **reconstruct the operational history of each sleeve** and prepare the data for lifecycle analysis and Formal Concept Analysis (FCA). First, the separate date and time fields were combined into a single timestamp to create a complete chronological record of the production process. The dataset was then sorted by sleeve identifier and timestamp so that all events belonging to the same sleeve could be analysed in their correct temporal order.

For each sleeve, a cumulative tonnage value was calculated by continuously summing the amount of processed steel. The maximum cumulative tonnage reached by a sleeve was considered its total operational capacity, and a new Remaining Useful Life (RUL) value was computed as the difference between this maximum capacity and the current cumulative tonnage.

To allow comparison between sleeves of different lifetimes, the calculated RUL was normalized into a **percentage scale** ranging from 100% at the beginning of the lifecycle to 0% at the end. Based on this, each observation was assigned to one of four health categories: Critical (0–25%), Low (25–50%), Medium (50–75%), and Healthy (75–100%).

Only the most relevant process, production and equipment attributes were retained for further analysis. To facilitate FCA and data mining, continuous numerical attributes were discretized into three categories (Low, Medium, High) and encoded into numerical representations. This transformation produced a dataset suitable for concept lattice construction, scale generation and lifecycle pattern discovery.

Table 2: Additional attributes generated during dataset preprocessing

Attribute	Description
datetime_combined	Combined date and time of continuous casting sample acquisition
calculated_RUL_tons	Estimated remaining useful life of the caster expressed in tons of steel processed
RUL_percentage	Remaining useful life expressed as a percentage of nominal lifetime
*_BIN	Discretized representation of continuous numerical variables (Low, Medium, High) used for rule extraction and interpretability
*_ENCODED	Integer-encoded representation of categorical and discretized variables for machine learning pipelines
steel_type, alloy_type, work-piece_slice_geometry	Categorical process descriptors of steel grade, alloy classification and slab geometry
sleeve, num_crystallizer, num_stream	Equipment configuration identifiers for the casting setup
temperature_measurements	Multiple temperature sensor readings used for process monitoring and validation
cooling_related variables	Includes water consumption, water temperature delta and derived cooling behavior descriptors
process_kinematics	Includes swing frequency, crystallizer movement and alloy speed describing caster dynamics
steel_weight, steel_temperature_grab1, resistance	Primary process measurements recorded during casting operations

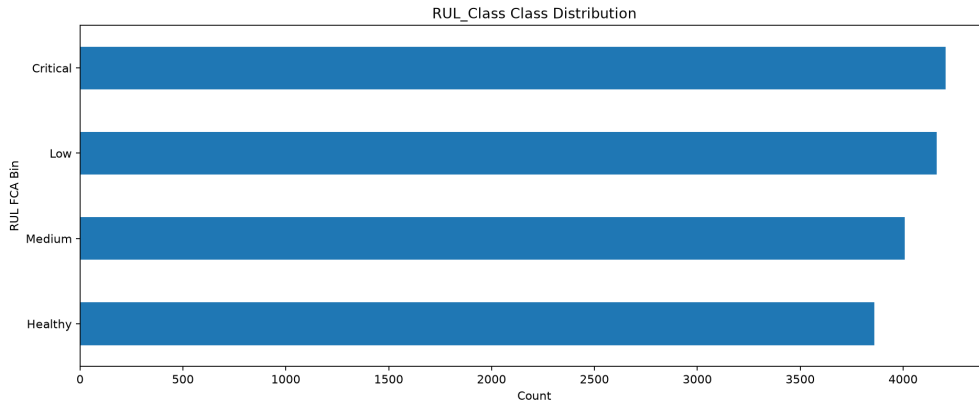


Fig. 6: RUL classes distribution

Also, the average RUL was analyzed based on different industrial groups, such as workpiece geometry, stream, crystallizer and alloy type. This helped in understanding how the RUL value changes across different production conditions.

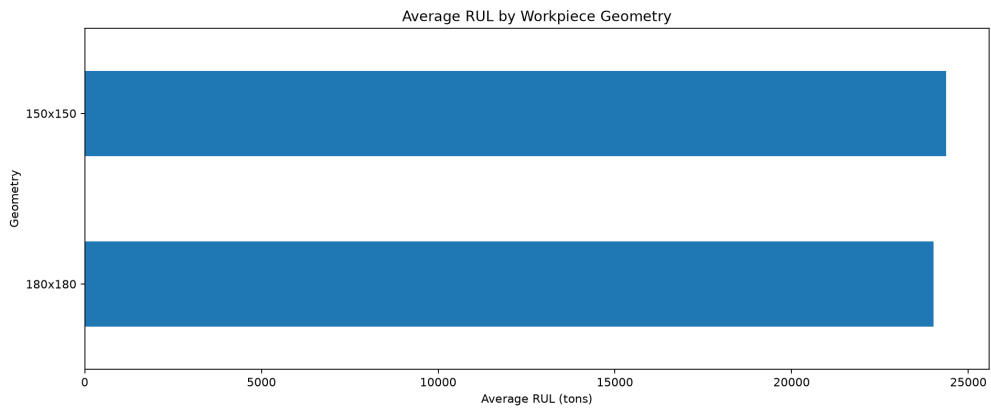


Fig. 7: Average RUL per geometry distribution

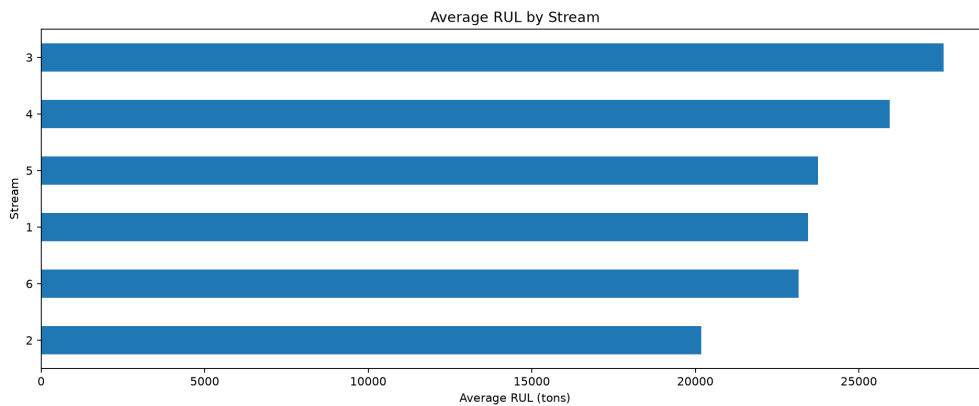


Fig. 8: Average RUL per stream distribution (equipment identifiers)

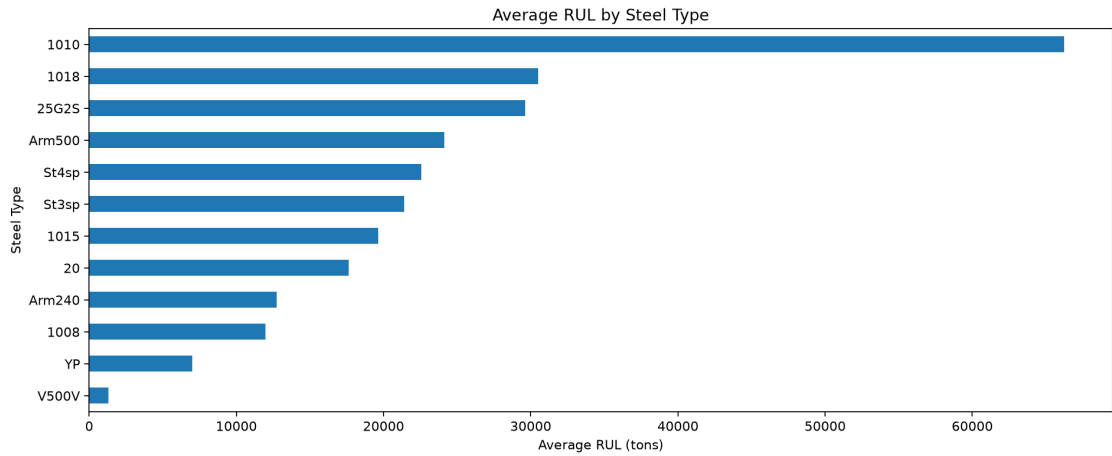


Fig. 9: Average RUL per steel type

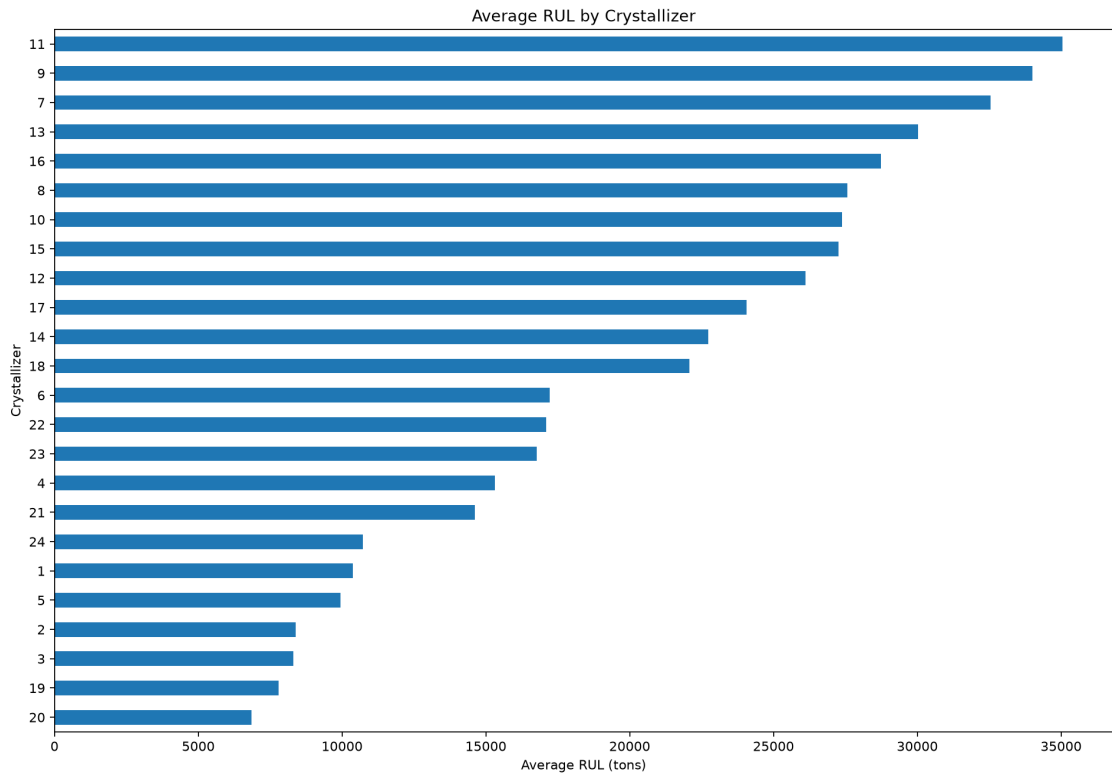


Fig. 10: Average RUL per crystallizer distribution (equipment identifiers)

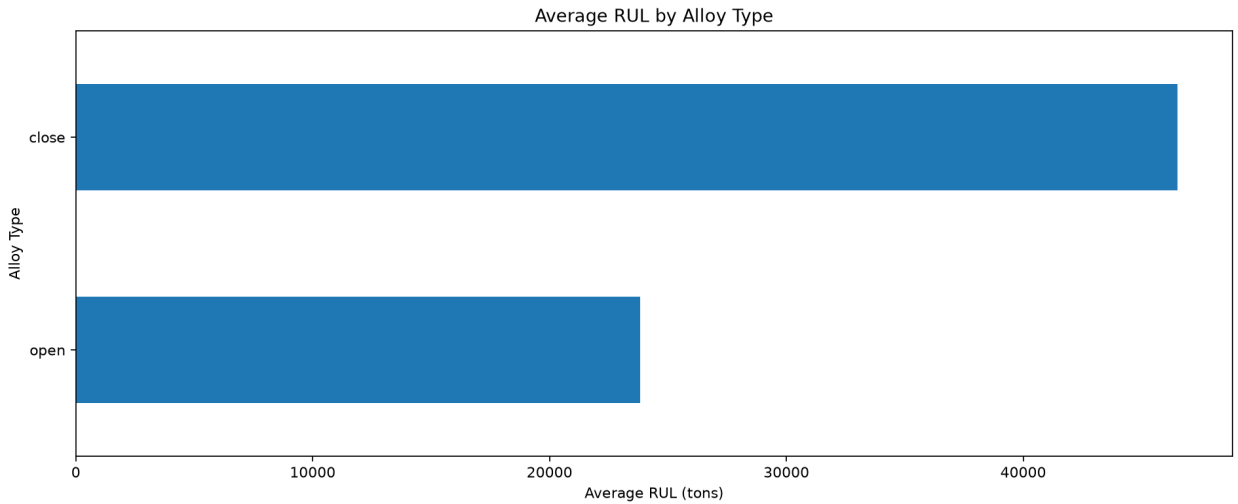


Fig. 11: Average RUL per alloy type

A histogram of the RUL distribution was generated to observe how the target values are spread.

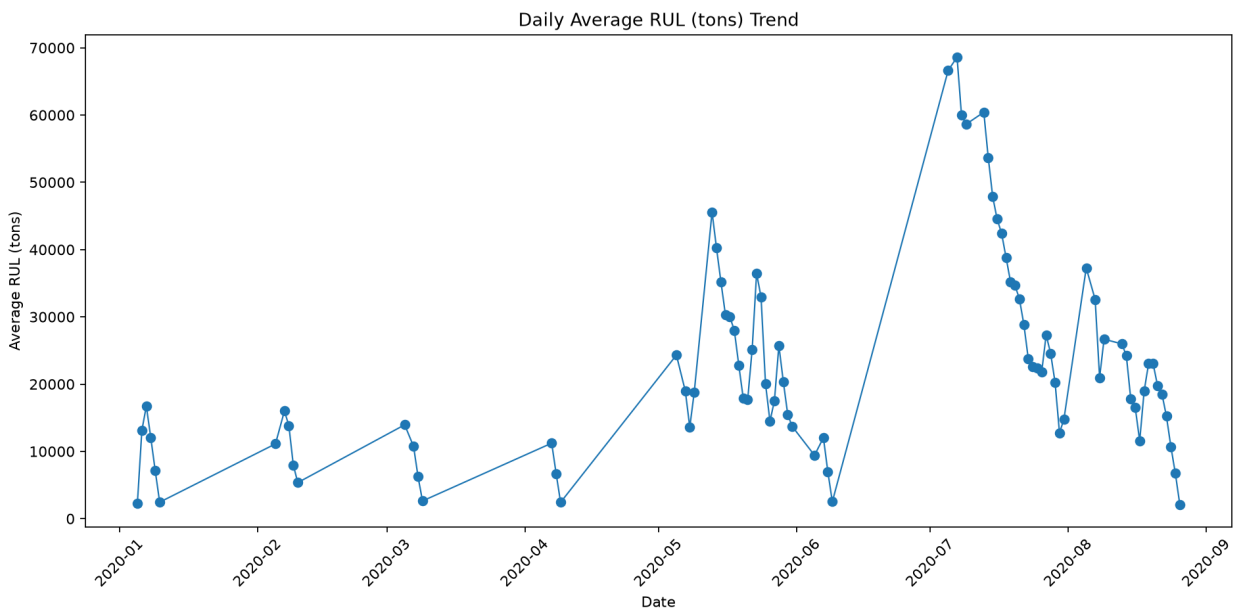
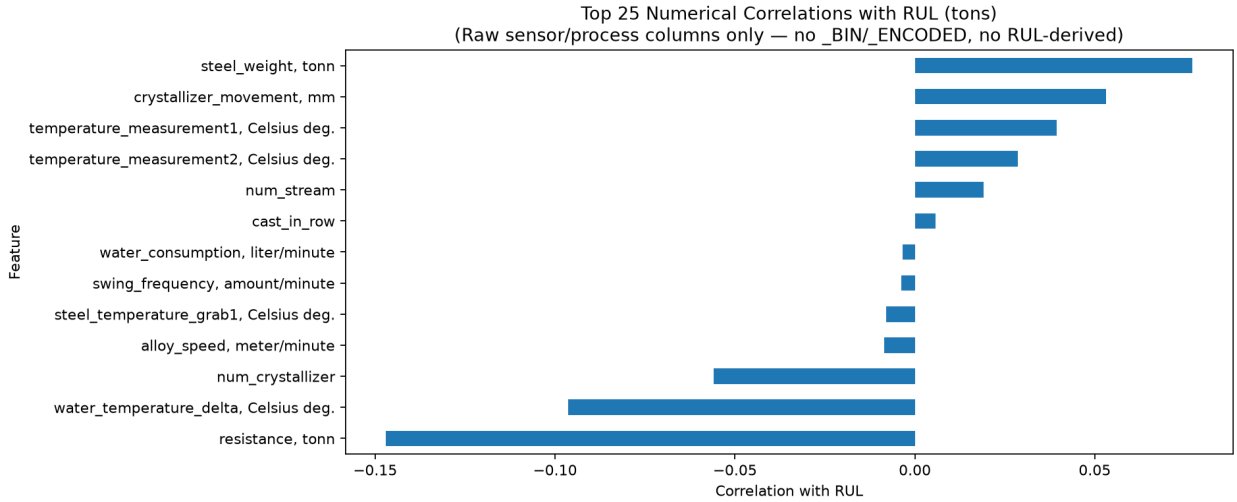


Fig. 12: Average RUL distribution history

Another important part of the exploratory analysis was the calculation of numerical correlations between the available features and the RUL target. This helped identify which variables are most strongly related to RUL. The following plot is used to visually check how sensor values change in relation to sleeve lifetime.

Fig. 13: Top numerical correlations **RUL** feature

4.2. Supervised learning

For supervised learning (**Supervised Learning (SL)**), **RUL** was used as a **regression target**. **SL** is a **machine learning approach** where a model is trained on **labeled data**, meaning each input example is paired with a known output, and the goal is to learn a mapping from inputs to outputs that can be used to predict the output for unseen data [3]. A **Random Forest Regressor (Random Forest Regressor (RFR))** was trained to predict **RUL** using process and sensor features. The **RFR** builds multiple decision trees on random subsets of the training data and features and the final prediction is obtained by averaging the outputs of all trees [4]. The dataset was split into training and testing sets and the model was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (rmse)** and **R-squared (R^2)**.

Metric	Value	Definition
R^2	0.90	Measures how well the model explains the variance in the target variable. A value close to 1 indicates a strong fit [11]
MAE	2748	The average absolute difference between predicted and actual values [5]
RMSE	5790	Square root of the average squared differences between predicted and actual values (penalizes large errors more strongly) [5]

Table 3: Evaluation metrics for the **RFR** model

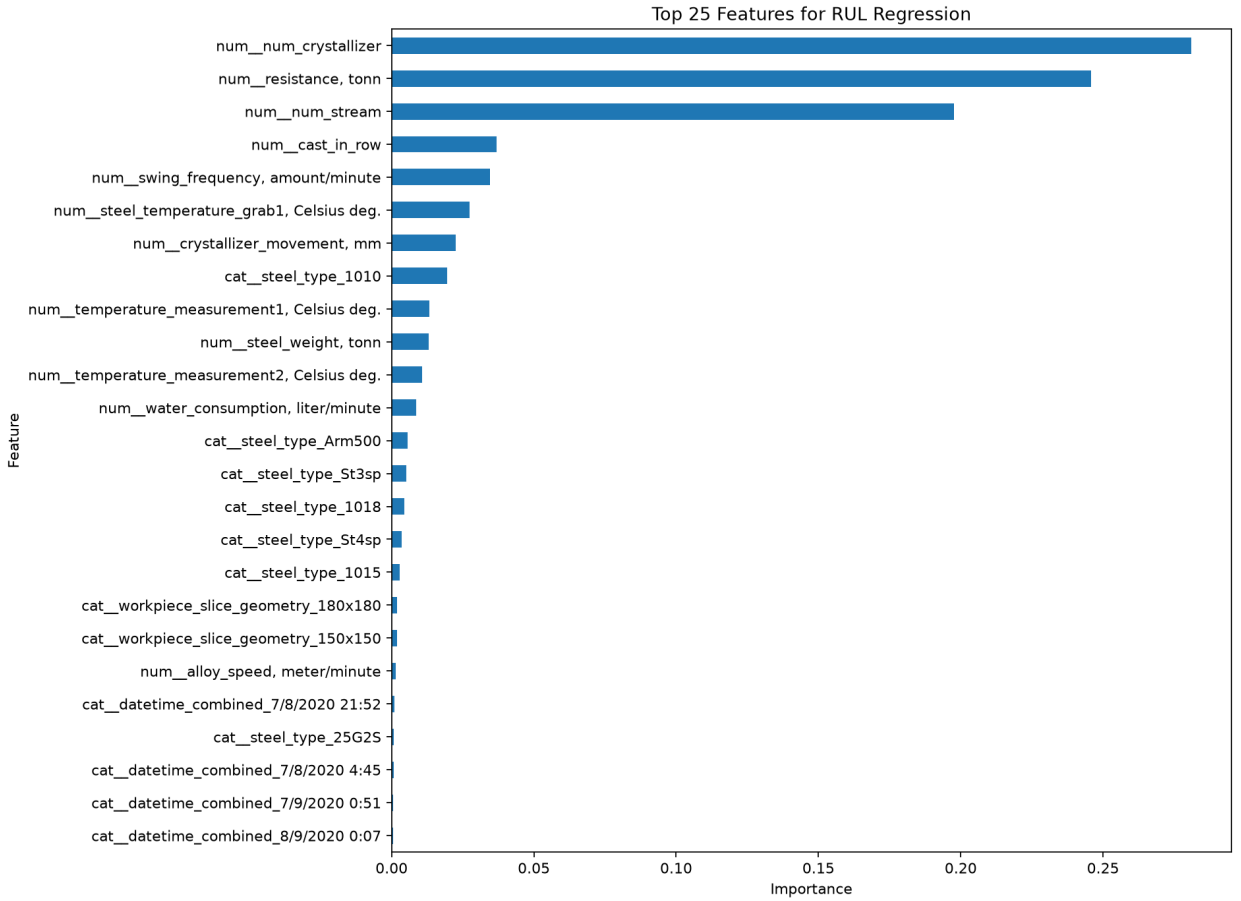


Fig. 14: Feature importance analysis for the RFR model

The previous plot determines which input features had the strongest effect on RUL predictions. It also makes the RFR model easier to interpret.

For classification, 'RUL_class' was used as the target variable. A **Random Forest Classifier** (*Random Forest Classifier (RFC)*) was trained to predict sleeve condition categories based on RUL classes. It constructs multiple decision trees on random subsets of the training data and features and the final class prediction is obtained by majority voting across all trees [4].

Metric	Value	Definition
Accuracy	0.6856	The proportion of correctly classified samples out of all predictions, measures overall correctness [15]
Precision (macro)	0.6865	The proportion of correctly predicted positive samples among all predicted positive samples [15]
Recall (macro)	0.6859	The proportion of correctly predicted positive samples among all actual positive samples [15]
F1-score (macro)	0.6862	The harmonic mean of precision and recall, providing a balanced measure between both metrics, especially useful for imbalanced datasets [15]

Table 4: Evaluation metrics for the RFC model

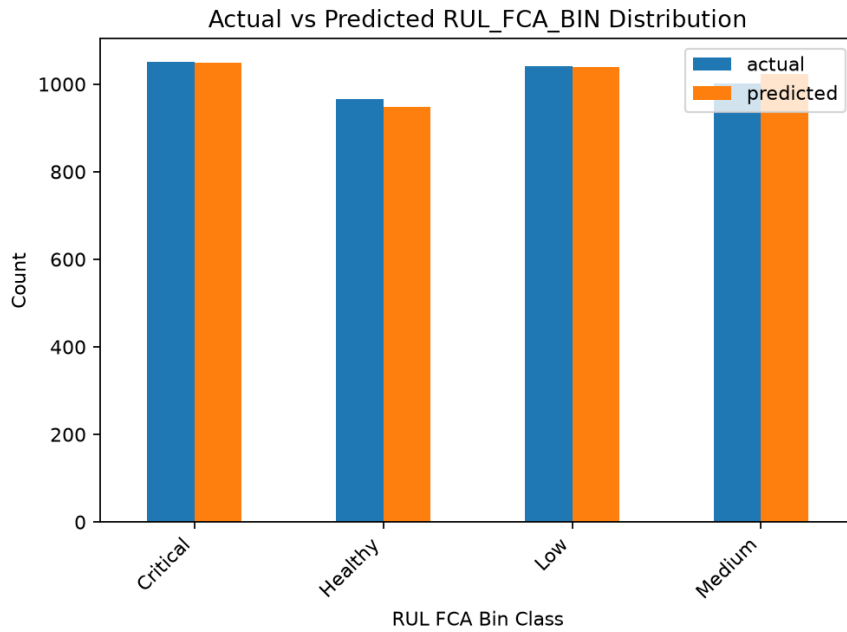


Fig. 15: Classification actual and predicted distributions for the **RFR** model

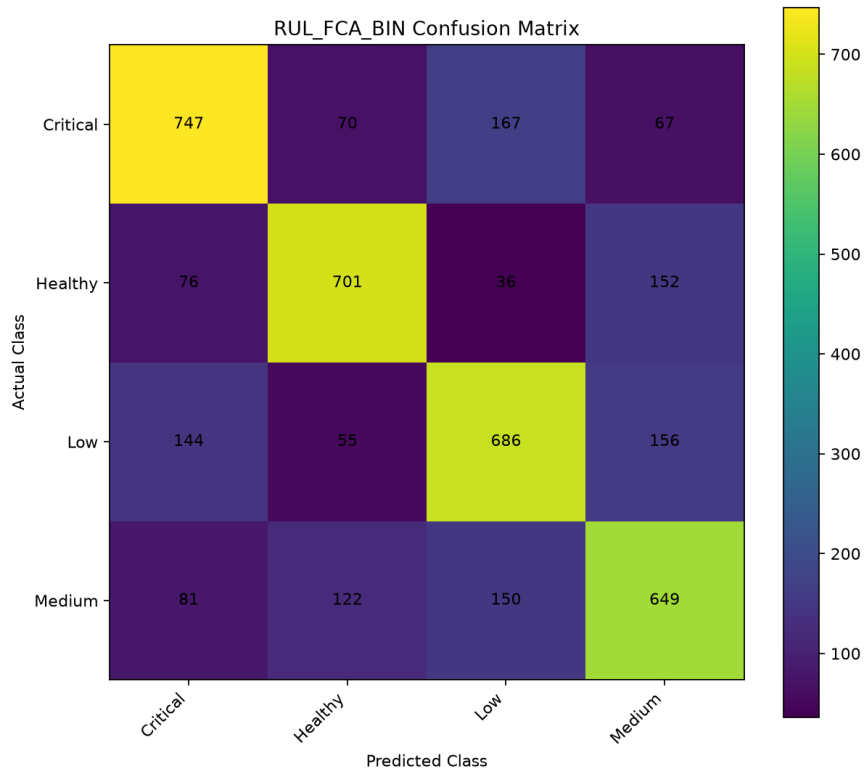


Fig. 16: Confusion matrix for the **RFR** model

4.3. Unsupervised learning

In addition to supervised learning (SL), some unsupervised methods were applied. **Unsupervised learning (Unsupervised Learning (UL))** is a ML approach that works with unlabelled data and Its goal is to discover hidden patterns, structures, relationships or groups within the data without using predefined target values [3]. **K-Means Clustering** was used to group similar production cases. K-Means is an unsupervised clustering algorithm that partitions data into a predefined number of clusters (K), in this case 4 (according the the 4 RUL classes defined) [3]. **Density-Based Spatial Clustering of Applications with Noise (Density-Based Spatial Clustering of Applications with Noise (DBSCAN))** was also applied to detect **dense clusters** and **noise points** [6]. **Principal Component Analysis (Principal Component Analysis (PCA))** was used to reduce the dataset to two main components for visualization and as a base for UL algorithms. PCA is a dimensionality reduction technique that transforms a dataset into a smaller set of uncorrelated variables called **principal components** [12]. These components capture most of the variance in the data, making it easier to visualize and analyse complex datasets while reducing information loss.

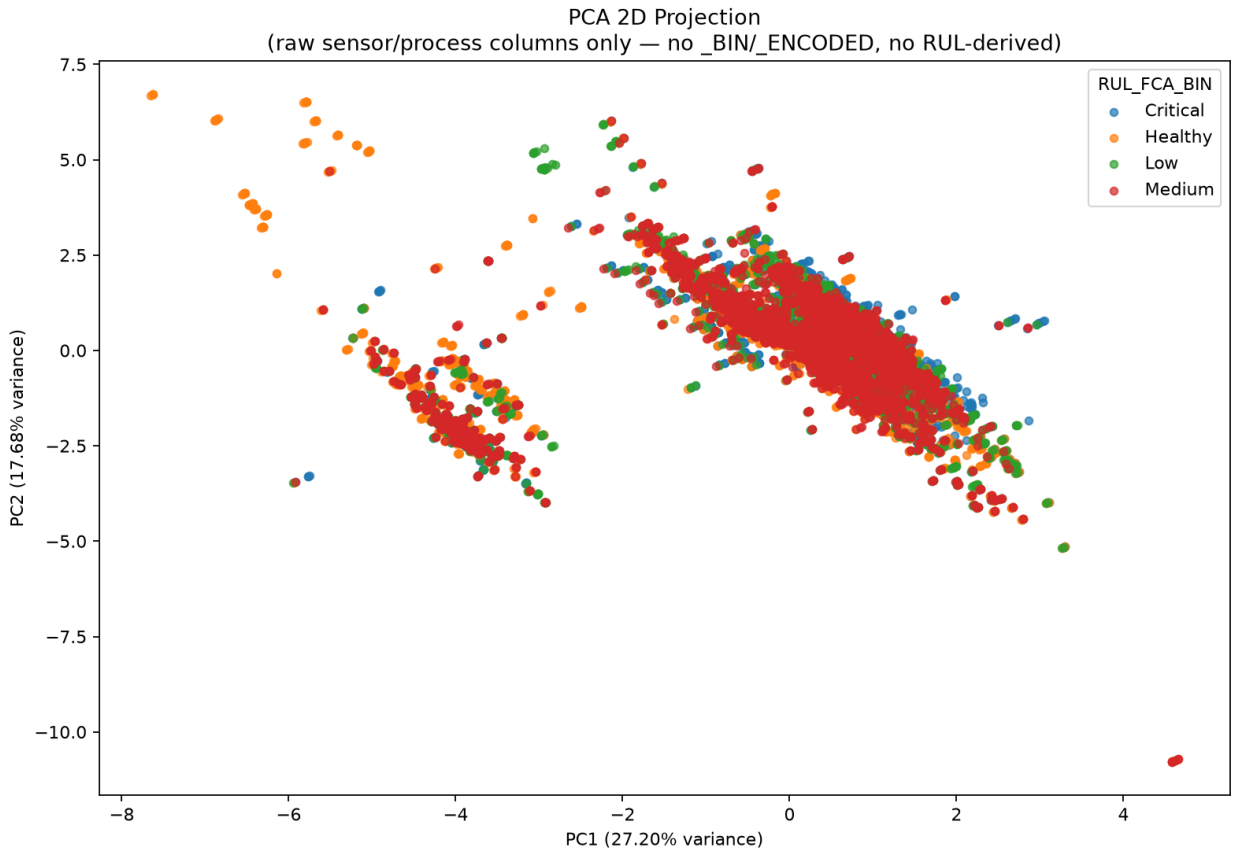


Fig. 17: PCA over RUL classes

Principal Component	Explained Variance Ratio
PC1	0.2720
PC2	0.1768

Table 5: Explained variance ratio of the first two principal components

The first principal component (*PC1*) explains 27.20% of the total variance, while the second principal component (*PC2*) explains 17.68%. Together, the two components account for 44.88% of the total variance. Although this captures a substantial portion of the dataset's variability, more than half of the variance remains outside the two-dimensional projection. Therefore, the **PCA visualization provides only a partial representation of the underlying data structure**. **K-Means** forces the data into a predefined number of clusters, which may not align with the intrinsic structure of the data. **DBSCAN** primarily groups most observations into one large cluster, with a smaller separate cluster, several tiny dense groups, and a small proportion of points classified as noise. Overall, these results suggest that a two-dimensional representation may not be sufficient to fully capture the complexity of the dataset and clustering approaches operating in higher dimensional feature spaces may provide more reliable results.

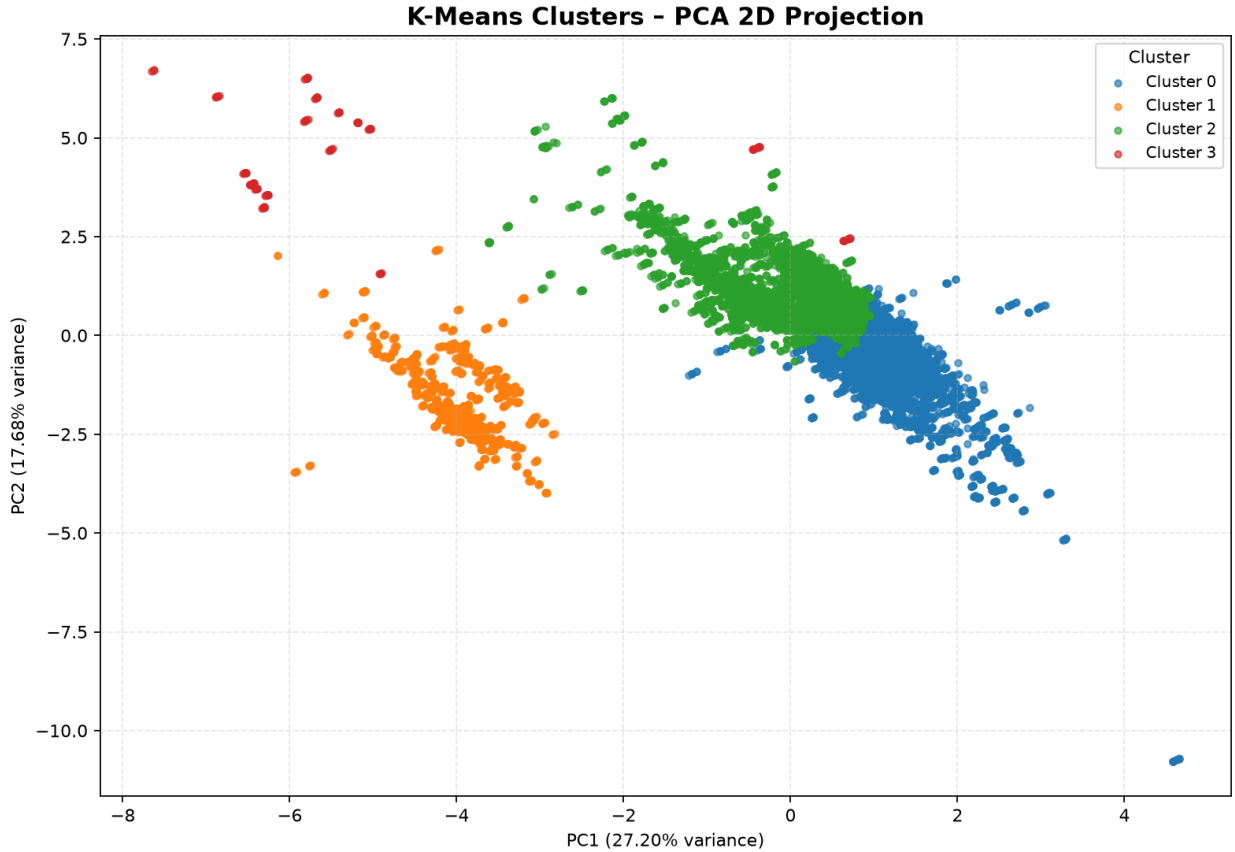


Fig. 18: K-Means Clustering over PCA

4.4. Anomaly Detection

An **Isolation Forest** (**Isolation Forest (IF)**) was used to find unusual production cases. **IF** is an unsupervised anomaly detection algorithm that identifies outliers by isolating data points through random feature splits [14].

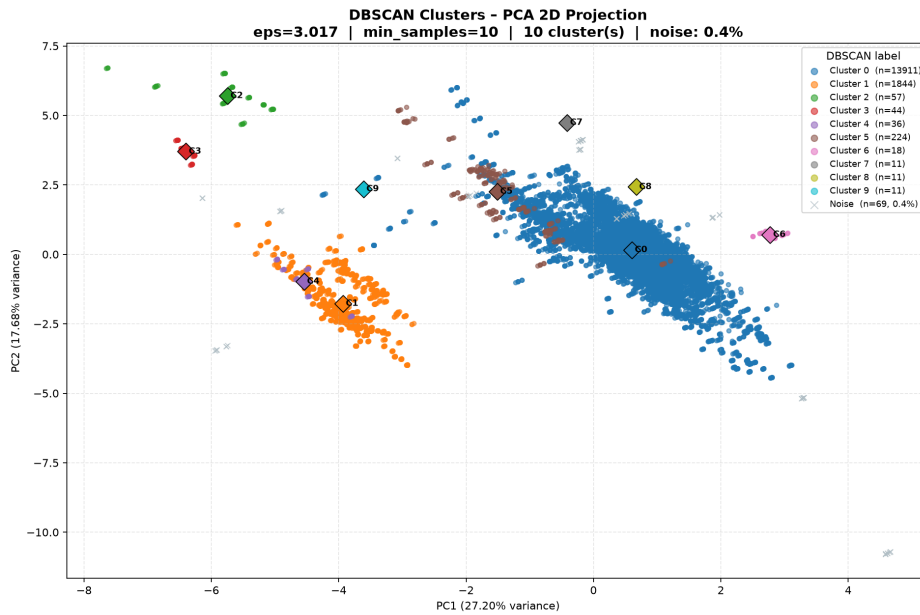


Fig. 19: **IF** anomaly over scatter **PCA**

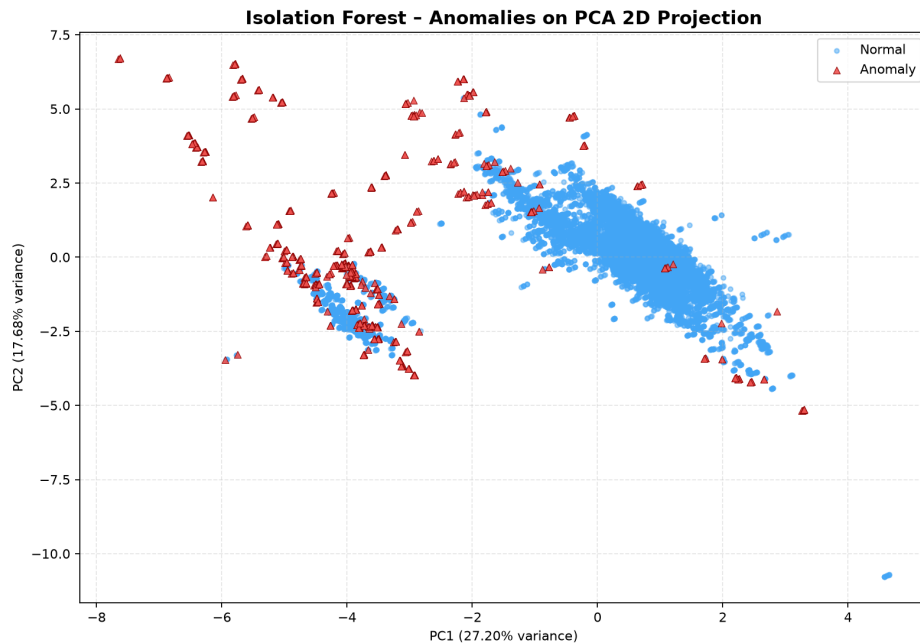


Fig. 20: **IF** anomaly heat map over **PCA**

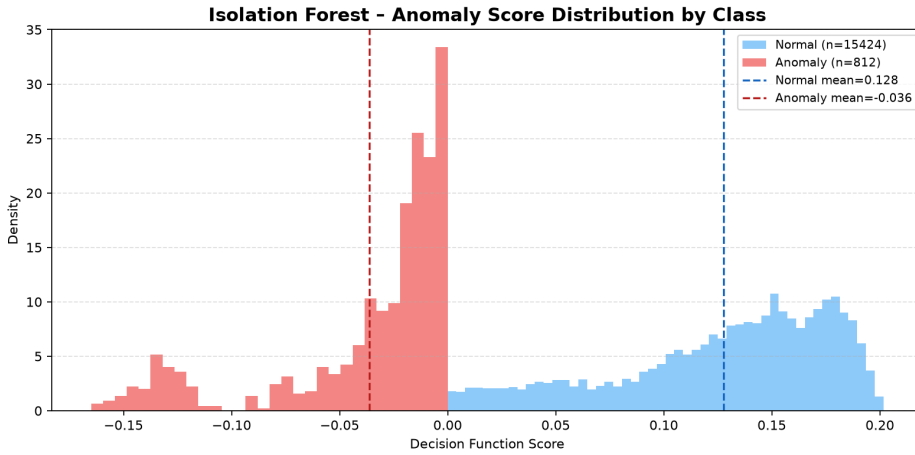


Fig. 21: IF score distribution

These cases may represent **abnormal behaviour** or **faster sleeve degradation**.

4.5. Association Rule Mining

Association rule mining was applied using Formal Concept Analysis [FCA](#) binary columns and the 'RUL_class' column. There, the **Apriori** algorithm was used. It is an unsupervised learning method used in association rule mining [1], used to find frequent item sets in large datasets and then generates rules that describe how items co-occur together. The resulting rules were ranked based on lift and confidence. **Confidence** measures the reliability of an association rule, indicates how often the rule is correct when the condition is satisfied. **Lift** measures the strength of an association rule compared to random chance, a lift value greater than 1 indicates a positive association, while a value close to 1 suggests no meaningful relationship.

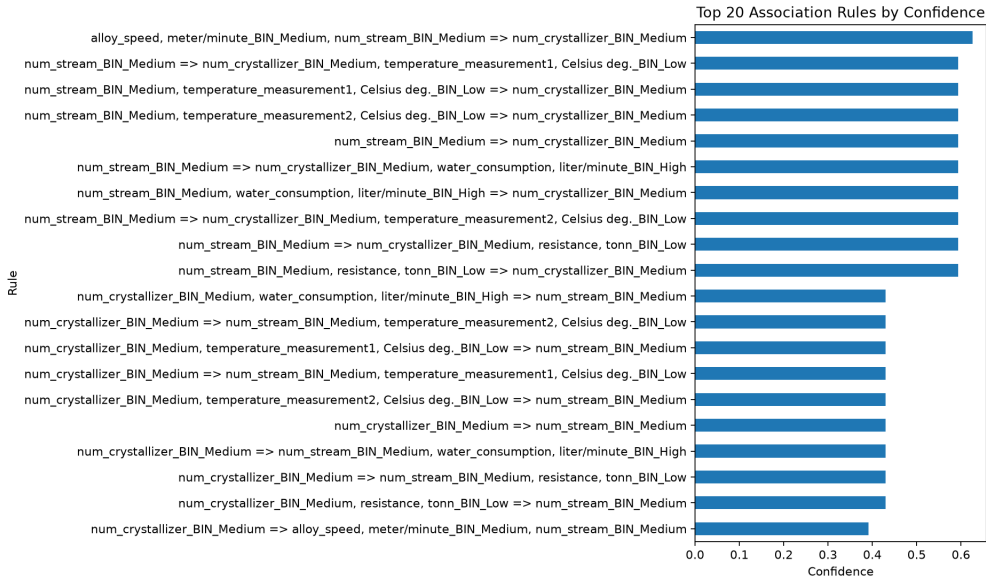


Fig. 22: Top 20 association rules by confidence

The top 20 rules ranked by confidence show high reliability, with values between 0.40 and 0.65. This means that when certain heavy material weights and high tonnage levels occur, the system is in a healthy **RUL** class in 40% to 65% of cases.

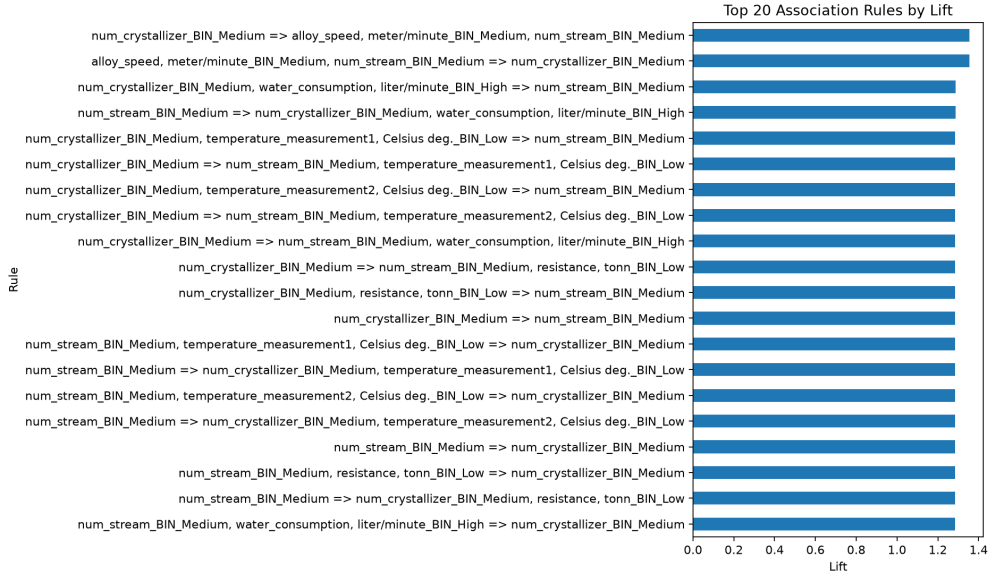


Fig. 23: Top 20 association rules by lift

The top 20 rules ranked by lift all have values above "1.2", reaching up to about "1.4". Since a lift greater than 1 indicates a real association beyond chance, these results show that these operating conditions are linked to machine behaviour.

As a conclusion for rule mining, **the results suggest that specific combinations of load and tonnage are good indicators of stable machine operation.**

5. Knowledge Discovery

5.1. Knowledge Discovery

First of all, let's define every field approached within this document. **Knowledge Discovery (KD)** is the process of extracting general, meaningful patterns from data rather than analysing isolated records [7]. Its objective is to transform raw data into structured and explainable knowledge that captures what is consistently true in the dataset.

KD focuses on identifying:

- **Patterns:** regular structures in data;
- **General rules:** valid beyond individual cases);
- **Stable relationships:** robust dependencies between variables.

5.2. Formal Concept Analysis

Formal Concept Analysis (**FCA**) is a mathematical framework for discovering structured knowledge in data [8]. It represents information as a set of **formal concepts**, each composed of:

- **Extent:** a set of objects;
- **Intent:** a set of shared attributes.